Consent in Crisis: The Rapid Decline of the AI Data Commons

Longitudinal audit of consent protocols for AI training data

NeurIPS (2024)

Web consent signals: robots.txt + Terms of Service

Relevant Conferences & Journals

CS / ML conferences

- NeurIPS (Dec)
- ICML (July)
- ICLR (April - May)
- AISTATS (Mar – Apr)
- AAAI (Jan - Feb)

Statistics journals

- *Journal of the American Statistical Association* (JASA)
- *Annals of Statistics*
- *JRSS-B* (Journal of the Royal Statistical Society, Series B)
- *Biometrika*

Quick Overview

- **Research question:** How is AI re-shaping the culture of web consent and shifting the landscape for AI training data?
- **Context:**
  - **Web crawler:** Automated bots that systematically browse and download web pages (e.g., search engines, Common Crawl).
  - **Robots Exclusion Protocol (REP):** A machine-readable file (`robots.txt`). It specifies which bots may access which URLs.
  - **Terms of Service (ToS):** Legal agreements written in natural language that define how website content may be accessed.
- **Findings:**
  - Increase in AI-related crawl restrictions.
  - Inconsistencies between robots.txt and ToS.
  - Different domain features.
  - Training data composition diverges from real-world usage.

Datasets

Three primary web-derived training corpora from Common Crawl

- **C4 (Colossal Clean Crawled Corpus)**
  - Cleaned, English-only text
- **RefinedWeb**
  - Large-scale filtered web corpus
  - Emphasizes scale over curated mixtures (books, Wikipedia, etc.)
- **Dolma**
  - Open 3T-token corpus
  - More structured, provenance-aware, governance-conscious

Sampling Strategy

- $HEAD_{All}$: Union of top 2k domains (by tokens) from each corpus $\Rightarrow$ 3.95k high-impact domains
- $HEAD_{C4}$: consider only the head of C4
- $RANDOM_{10k}$: 10k domains randomly sampled from intersection
- $RANDOM_{2k}$: random 2k subset of $RANDOM_{10k}$ for human annotation

Datasets: Human Annotations

ATTRIBUTE	DETAILS	COLLECT
Content Modalities	Whether the web domain has images, videos, and standalone audio in addition to text.	✒️
User Content	Whether the web domain hosts primarily content provided by users, such as forums, blog hosting, and social media websites.	✒️
Sensitive Content	Whether explicit, illicit, pornographic, or hate speech content is clearly present.	✒️
Paywall	Whether the web domain has use limits or any access gating behind a paywall.	✒️
Advertisements	Whether the web domain has automatic advertisements embedded into any of its pages.	✒️
Purpose & Service	The purpose or service(s) of a website? Options: E-commerce, Social Media/Forum, Encyclopedia, Academic, Government, Organization site, News, or Other.	✒️
	Terms & Restrictions	
Robots.txt	A web domain's robots.txt restrictions on crawler agents. We use Google's crawler rules.	🤖 ⏱️
Terms & Policies	The terms, content, copyright, and privacy policy pages found for a web domain.	✒️ ⏱️
Crawling & AI Policy	Do terms restrict both crawling and AI, restrict crawling, restrict only AI, conditionally restricting crawling/AI, or not apply restrictions?	🤖 ⏱️
Content Use Policy	Are there content use restrictions. Options: restricted to personal, academic, or non-commercial use, conditionally restricted, or unrestricted.	🤖 ⏱️
Non-Compete Policy	Is content use prohibited for developing competing services?	🤖 ⏱️

Table 2: The **list of attributes collected for each web domain**, as sampled from C4, Dolma, and RefinedWeb. 🤖 denotes automatic collection; ✒️ denotes human annotation; ⏱️ denotes this information was collected historically from 2016, as well as statically. Full annotation guidelines are given in Appendix A.2.2.

Datasets: Longitudinal Consent Audit

From cross-section domain to panel

- Monthly snapshots from **January 2016 to April 2024**
- Source: **Wayback Machine** (835B+ archived web pages)
- Collected for each domain:
  - Homepage
  - `robots.txt` (REP file)
  - Terms of Service / policy pages

Crawlers analyzed

- AI developers: Google, OpenAI, Anthropic, Cohere, Meta
- Non-profit archives: Common Crawl, Internet Archive

Key outcome: Restricted Tokens at domain level

- **Robots.txt restriction:**

  % tokens from domains that fully block at least one AI's all crawlers

- **ToS restriction:**

  % tokens unusable due to legal clauses restricting crawling or AI

- Robots.txt redistributed in 2023, introduction of GPTBot and Google-Extended crawler agents.

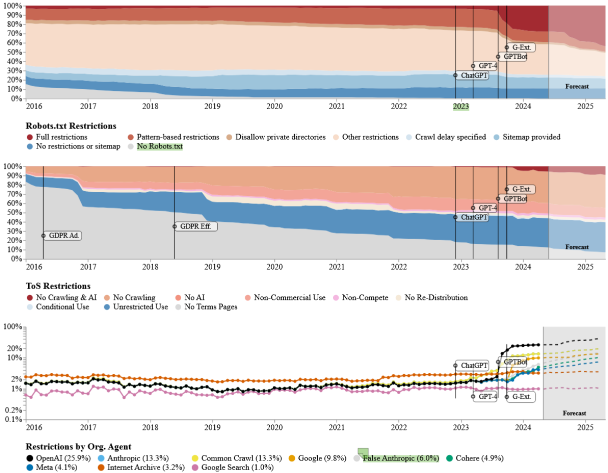
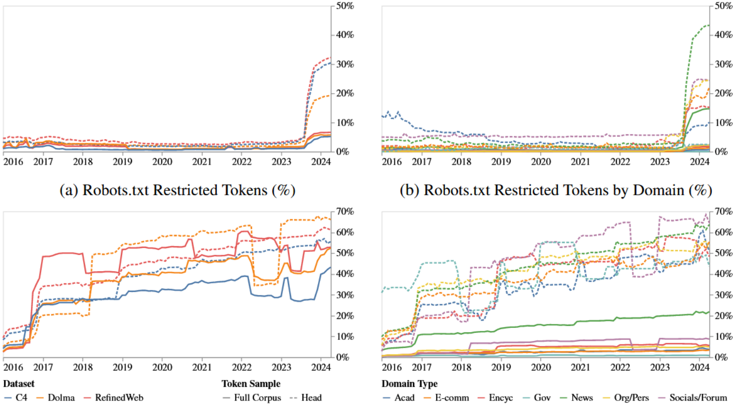


Figure 1: A temporal analysis, from 2016 to April 2024, of the web consent signals in HEAD$_{C4}$, a

Finding 1: The Rise of Restrictions on Open Web Data

- AI restrictions are driven by news, forums, and social media.



(a) Robots.txt Restricted Tokens (%)

(b) Robots.txt Restricted Tokens by Domain (%)

(c) Terms of Service Restricted Tokens (%)

(d) Terms of Service Restricted Tokens by Domain (%)

Figure 2: A temporal depiction of the percentage of restricted tokens across both the Full Corpus and the HEAD$_{All}$ sample—the largest and most critical data sources. The robots.txt analysis (top) and

Finding 2: Inconsistent Communication on AI Consent

- Some AI crawlers are allowed, while others are not.
- Unrecognized crawler agents cause incorrect specifications.
- Contradictions exist between robots.txt and ToS.

Robots Restrictions \ Terms of Service Policies	None	Conditional	No Distribution	Non-Compete	NC Only	No AI	No Crawling	No Crawling or AI
Restricted	6.3 %	0.3 %	0.3 %	0.4 %	2.3 %	--	8.6 %	2.7 %
Partial	14.0 %	0.2 %	1.8 %	1.6 %	5.1 %	0.1 %	12.4 %	0.9 %
None	5.6 %	0.1 %	0.3 %	0.1 %	1.9 %	--	34.9 %	0.2 %

Figure 3: A cross tabulation of the Terms of Service policies and robots.txt restrictions for HEAD_{C4}, measured in percentage of tokens. **We find two ways of expressing restrictions on data use for AI *frequently disagree***, both in what they express, and can express.

Organization	Rest. (%)
OpenAI	91.5
Common Crawl	83.4
Anthropic	83.4
Google Extended	72.0
False Anthropic	61.6
Cohere	52.3
Meta	52.2
Internet Archive	32.3
Google Search	17.1

Table 3: The % each org's crawler agents are **restricted** if at least one other org in this pool is restricted. Gray indicates crawlers with a primary purpose other than AI training data.

Finding 2: Inconsistent Communication on AI Consent

Organization	User Agent	Details	Docs	Purpose
OpenAI	GPTBot	OpenAI's official user-agent for crawling and collecting training data from the web.	●	Training
	ChatGPT-User	OpenAI's official user-agent for live user queries that trigger browsing plugins. According to OpenAI's documentation, their current opt-out implementation treats both user agents the same.	●	Retrieval
Google	Google-Extended	Google's official user-agent for "Gemini Apps, Vertex AI generative APIs, and future generations of models."	●	Training
Google Search	Googlebot	Google's official user-agents for general web crawling, related to their search engine.	●	Web Search
Common Crawl	CCBot	Common Crawl's user-agent for maintaining open access archives of the web, particularly for research.	●	Archive, research
Anthropic	ClaudeBot	Anthropic's official user-agent for crawling and collecting training data from the web. Their policy statest that their opt-outs respect this agent as well as Common Crawl's CCBot.	●	Training, retrieval
False Anthropic	anthropic-ai	An unofficial but widely adopted user-agent, presumably to disallow any Anthropic data crawling and collection.	○	Training
	Claude-Web	An unofficial but widely adopted user-agent, presumably for live queries in Claude which trigger browsing.	○	Retrieval
Meta	FacebookBot	Meta's official user-agent for crawling and collecting training data from the web	●	Training, retrieval
Cohere	cohere-ai	An unofficial but widely adopted user-agent, presumably to disallow any Cohere data crawling and collection.	○	Training, retrieval
Internet Archive	ia_archiver	The official user-agent that supports the Wayback Machine open web archive. The Internet Archive may ignore this user-agent		Training, retrieval
All Agents	*All*	Notation for our aggregation of robots.txt policies towards all agents. This is used to track if a website is fully lenient or restrictive to all user agents.		

Table 5: The **list of organizations we trace, and their associated web crawler user-agents.**

Finding 2: Inconsistent Communication on AI Consent

Agent Name	# Observed	All Disallowed		Some Disallowed		None Disallowed	
		Count	%	Count	%	Count	%
All Agents	269,212	1175	0.44%	198642	73.79%	69395	25.78%
*	226,903	2935	1.29%	183364	80.81%	40604	17.89%
Mediapartners-Google	20,848	749	3.59%	3710	17.80%	16389	78.61%
Googlebot	12,831	44	0.34%	9544	74.38%	3243	25.27%
MJ12bot	10,556	5962	56.48%	319	3.02%	4275	40.50%
Twitterbot	10,385	29	0.28%	3413	32.86%	6943	66.86%
Slurp	10,070	507	5.03%	4364	43.34%	5199	51.63%
AhrefsBot	9,824	5506	56.05%	1516	15.43%	2802	28.52%
IRLbot	9,142	134	1.47%	157	1.72%	8851	96.82%
Yandex	8,948	2901	32.42%	1623	18.14%	4424	49.44%
bingbot	8,077	195	2.41%	3079	38.12%	4803	59.47%
Googlebot-News	7,958	82	1.03%	7548	94.85%	328	4.12%
Baiduspider	7,789	3476	44.63%	1933	24.82%	2380	30.56%
msnbot	6,784	261	3.85%	2737	40.34%	3786	55.81%
008	5,661	5360	94.68%	108	1.91%	193	3.41%
ia_archiver	5,604	2768	49.39%	1822	32.51%	1014	18.09%
SemrushBot	5,418	3865	71.34%	84	1.55%	1469	27.11%
Googlebot-Image	5,082	728	14.33%	1842	36.25%	2512	49.43%
Nutch	5,011	4446	88.72%	0	0.00%	565	11.28%
ccBot	5,005	3227	64.48%	1328	26.53%	450	8.99%
facebookexternalhit	2,286	31	1.36%	323	14.13%	1932	84.51%
NPBot	2,261	2259	99.91%	0	0.00%	2	0.09%
wget	2,234	2234	100.00%	0	0.00%	0	0.00%
rogerbot	2,147	736	34.28%	703	32.74%	708	32.98%
libwww	2,138	2046	95.70%	0	0.00%	92	4.30%
SemrushBot-SA	2,108	1428	67.74%	10	0.47%	670	31.78%
sitecheck.internetseer.com	2,064	1972	95.54%	0	0.00%	92	4.46%
Download Ninja	2,060	1971	95.68%	0	0.00%	89	4.32%
ZyBORG	2,059	1941	94.27%	0	0.00%	118	5.73%
Zealbot	2,058	1969	95.68%	0	0.00%	89	4.32%
Xenu	2,048	1959	95.65%	0	0.00%	89	4.35%
Facebot	2,020	0	0.00%	936	46.34%	1084	53.66%
linko	1,991	1902	95.53%	0	0.00%	89	4.47%
ChatGPT-User	895	750	83.80%	118	13.18%	27	3.02%
anthropic-ai	260	229	88.08%	1	0.38%	30	11.54%
cohere-ai	185	180	97.30%	1	0.54%	4	2.16%
Google-Extended	871	836	95.98%	4	0.46%	31	3.56%
Amazonbot	546	358	65.57%	109	19.96%	79	14.47%
FacebookBot	235	220	93.62%	2	0.85%	13	5.53%
ClaudeBot	45	40	88.89%	0	0.00%	5	11.11%
Claude-Web	89	82	92.13%	0	0.00%	7	7.87%

Finding 3: What does Web Data Look Like?

Variable	URL Group				Stats	Pct. Tokens in Corpus		
	Top 100	Top 500	Top 2000	Random	Diff	C4	RW	Dolma
Restrictive Robots.txt	**38.4**	**35.0**	**26.5**	3.4	+23.1	5.0±1.5	6.6±2.3	5.6±1.9
Restrictive Terms	**64.1**	**61.0**	**51.2**	15.7	+35.5	43.2±15.2	52.8±30.3	52.3±15.4
User Content	**21.3**	19.1	**19.4**	15.1	+4.4	27.9±12.3	39.8±32.8	37.3±16.7
Paywall	**31.8**	**31.3**	**24.6**	1.6	+23.0	4.1±1.1	4.9±0.4	10.8±1.2
Ads	**54.6**	**61.4**	**53.2**	5.4	+47.9	23.5±12.6	44.8±34.4	34.8±18.1
Modality: Image	96.8	97.0	96.7	95.0	+1.7	97.7±2.3	98.6±0.9	97.5±1.9
Modality: Video	**87.0**	**78.8**	**58.7**	18.9	+39.8	32.9±14.2	27.0±14.7	35.4±10.6
Modality: Audio	**80.7**	**68.3**	**41.8**	3.4	+38.4	21.2±14.7	12.5±6.3	20.5±6.7
Sensitive Content	0.0	0.4	1.1	0.6	+0.5	0.8±1.0	0.2±0.4	1.8±3.0
Web Domain Service & Purpose								
Academic	**14.1**	**10.1**	**9.8**	3.8	+6.0	3.1±1.6	2.6±1.2	3.0±0.7
Blogs	**2.6**	**2.9**	**3.9**	15.1	-11.2	23.2±11.3	16.3±16.0	20.1±11.9
E-Commerce	8.4	9.9	10.1	10.6	-0.5	20.0±17.8	32.6±37.6	17.7±19.1
Encyclopedia/Database	**20.5**	**13.2**	**11.1**	0.4	+10.7	3.5±3.4	5.8±9.8	5.1±5.8
Government	**3.2**	**2.8**	**2.8**	1.1	+1.7	0.9±0.9	0.9±0.8	0.8±0.6
News/Periodicals	**45.6**	**53.3**	**50.0**	5.3	+44.7	11.5±3.9	16.8±10.8	22.9±10.9
Org/Personal Website	**15.3**	**13.2**	**12.7**	71.2	-58.5	48.5±13.3	57.3±24.2	46.3±14.2
Social Media/Forums	**9.4**	**9.3**	**11.8**	1.6	+10.1	5.1±4.8	5.4±8.9	14.9±8.3
Other	**15.0**	**10.9**	**11.8**	4.3	+7.4	4.7±2.7	2.8±1.3	3.7±2.0

Table 4: **Mean incidence rates of web source features across C4, RefinedWeb, and Dolma.** We measure incidence rates for the top 100, 500, and 2000 URLs, ranked by number of tokens, as well

Finding 4: Misalignment between Real-world AI Usage and Web Data

WildChat Data Construction

- WildChat: 1M real ChatGPT conversations collected via HuggingFace wrapper.
- Authors manually clustered 100 conversations by task intent.
- GPT-4o used to label 1k conversations using same taxonomy as web domains.
- Time window: Apr 9, 2023 – May 1, 2024 (GPT-3.5-Turbo, GPT-4).
- Validated by trained annotators.

Caveats

- User: Hosted on HuggingFace → potentially more technical users.
- Usage distributions limited to certain time periods and LLM models.

Finding 4: Misalignment between Real-world AI Usage and Web Data

- Web training sources heavily dominated by news, encyclopedias.
- Real-world usage dominated by creative writing, reasoning, coding.
- Sexual content and creative composition underrepresented in training corpora.

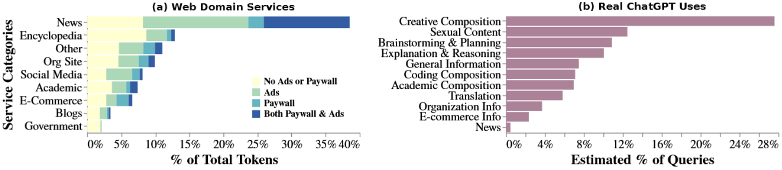


Figure 4: **The most common services provided by web domains in HEAD$_{C4}$ do not match real ChatGPT use cases from WildChat user logs.** Left: We measure the proportion of tokens in HEAD$_{C4}$ dedicated to each type of web service, and the degree to which they are monetized via paywalls and ads. Right: We measure the proportion of each type of user query in WildChat.

Conclusion

An important emerging phenomenon

- The “AI data commons” is not stable.
- Web consent norms are rapidly evolving in response to AI training.
- Data supply for foundation models is becoming endogenous.

Key insight

- High-quality, monetized, and frequently maintained domains (news, forums, paywalled content) are more likely to restrict AI crawlers.
- Restrictions disproportionately affect the most token-dense sources.

Broader implication

- Data is economically valuable — and increasingly strategically protected.
- The future of AI scaling may depend as much on data governance as on model architecture or compute.